

EXPERIMENT NO: 08

Experiment Title: Access and validate form fields using events; design a gym admission form with live input checks.

Software/Tools Required:

- Any text editor / VS Code
- Web Browser (Chrome/Edge) with Developer Console

Theory:

DOM Form Handling & Events: Form fields in HTML can be accessed through JavaScript using DOM selector methods (`document.getElementById()`, `document.querySelector()`). Event listeners allow programs to react dynamically as users interact with form controls.

- **Key Form Events:**
 - `input`: Triggers immediately when the value of an `input` element changes (ideal for real-time/live validation).
 - `blur`: Fires when an element loses focus (useful for field-level validation after the user finishes typing).
 - `change`: Triggers when an element's value is committed (e.g., selecting an option in a dropdown or checking a box).
 - `submit`: Fires when a form is submitted. Using `event.preventDefault()` stops the default page reload, allowing client-side validation.

Input Validation: Form validation ensures user inputs conform to expected formats before sending data to a server. Combining regular expressions (`test()`), conditional logic, and live DOM updates provides instant visual feedback to users, enhancing usability and reducing input errors.

Experiment Program Code

Code 1: Change.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
</head>
<body>
  <h3>Select Your Courses</h3>
  <select id = "course">
    <option values = "">Select Course</option>
    <option values = "CSE">CSE</option>
    <option values = "IT">IT</option>
    <option values = "ECE">ECE</option>
  </select>

  <p id = "result"></p>

  <script>
    let select = document.getElementById("course");
```

```

        select.addEventListener("change", function(){
            document.getElementById("result").innerHTML = "You selected: "+select.value;
        });
    </script>
</body>
</html>

```

Code 2: Submit.html

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <form id = "myform">
        <input type = "text" id = "name" placeholder = "Enter your name">
        <button type = "submit">Submit</button>

    </form>

    <p id = "result"></p>

    <script>
        let form = document.getElementById("myform");
        form.addEventListener("submit", function(event){
            event.preventDefault();
            let name = document.getElementById("name").value;

            document.getElementById("name").value;

            document.getElementById("result").innerHTML = "Hello " + name;

        });
    </script>
</body>
</html>

```

Code 3: EventListener.html

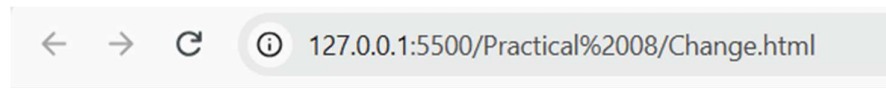
```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
</head>
<body>
    <button id = "btn">Click Me</button>

    <script>
        document.getElementById("btn").addEventListener("click", function(){
            alert("Button Clicked");
        });
    </script>
</body>
</html>

```

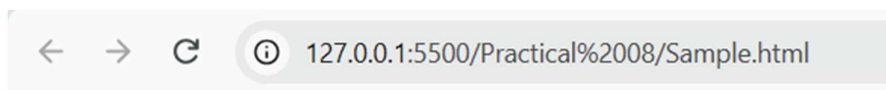
4. Output



Select Your Courses

CSE

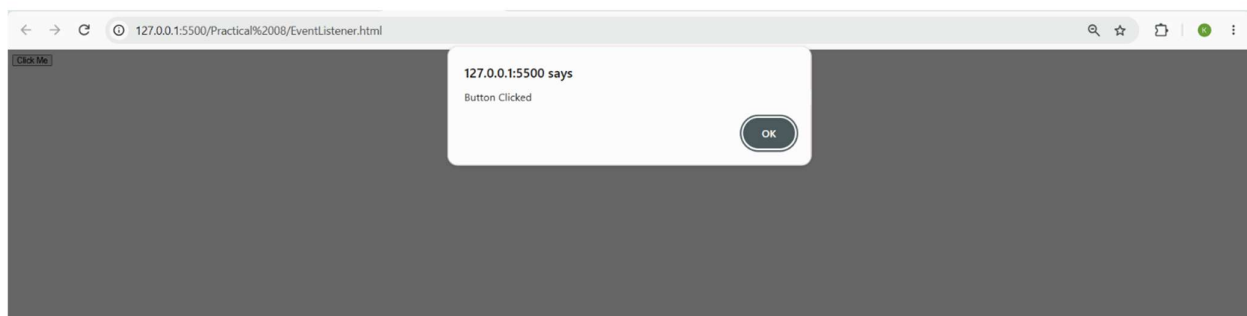
You selected: CSE



Enter Your Name

Kanak Tembhurne

Hello, Kanak Tembhurne



5. Case Study Title

Design a gym admission form with live input checks.

6. Case Study Program Code

Casestudy_08_Kanak_Tembhurne_1290.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Gym Admission Form</title>
```

```

<style>
  * {
    box-sizing: border-box;
    margin: 0;
    padding: 0;
  }

  body {
    font-family: Arial, sans-serif;
    background: #f4f1ea;
    min-height: 100vh;
    padding: 30px;
  }

  .container {
    width: 100%;
    max-width: 750px;
    margin: auto;
    background: white;
    padding: 35px;
    border-radius: 15px;
    box-shadow: 0 8px 25px rgba(0, 0, 0, 0.12);
  }

  h1 {
    text-align: center;
    color: #333;
    margin-bottom: 8px;
  }

  .subtitle {
    text-align: center;
    color: #777;
    margin-bottom: 30px;
  }

  .form-group {
    margin-bottom: 20px;
  }

  label {
    display: block;
    margin-bottom: 7px;
    font-weight: bold;
    color: #333;
  }

  input,
  select {
    width: 100%;
    padding: 12px;
    border: 2px solid #ddd;
    border-radius: 8px;
    font-size: 15px;
    outline: none;
  }

  input:focus,
  select:focus {
    border-color: #555;
  }

  .gender-options {
    display: flex;
    gap: 25px;
    margin-top: 8px;
  }

  .gender-options label {
    font-weight: normal;
  }

  .gender-options input {
    width: auto;
    margin-right: 5px;
  }

```

```

.message {
  font-size: 13px;
  margin-top: 5px;
  min-height: 18px;
}

.success {
  color: green;
}

.error {
  color: red;
}

.valid-input {
  border-color: green;
}

.invalid-input {
  border-color: red;
}

button {
  width: 100%;
  padding: 13px;
  border: none;
  border-radius: 8px;
  background: #333;
  color: white;
  font-size: 16px;
  cursor: pointer;
  margin-top: 10px;
}

button:hover {
  background: #555;
}

/* Registration Summary */

#registrationResult {
  display: none;
  margin-top: 30px;
  border: 2px solid #ddd;
  border-radius: 12px;
  overflow: hidden;
}

.result-header {
  background: #333;
  color: white;
  padding: 16px;
  text-align: center;
}

.result-header h2 {
  margin-bottom: 5px;
}

.result-header p {
  font-size: 14px;
}

.details {
  padding: 20px;
}

.detail-row {
  display: flex;
  justify-content: space-between;
  gap: 20px;
  padding: 12px 5px;
  border-bottom: 1px solid #eee;
}

.detail-row:last-child {

```

```

    border-bottom: none;
  }

  .detail-label {
    font-weight: bold;
    color: #555;
  }

  .detail-value {
    color: #222;
    text-align: right;
  }

  .registered {
    margin-top: 15px;
    padding: 12px;
    background: #e8f5e9;
    color: #2e7d32;
    border-radius: 8px;
    text-align: center;
    font-weight: bold;
  }

  @media (max-width: 600px) {
    .container {
      padding: 25px 20px;
    }

    .gender-options {
      flex-direction: column;
      gap: 8px;
    }

    .detail-row {
      flex-direction: column;
      gap: 5px;
    }

    .detail-value {
      text-align: left;
    }
  }
</style>
</head>

<body>

  <div class="container">

    <h1>Gym Admission Form</h1>

    <p class="subtitle">
      Register today and start your fitness journey
    </p>

    <form id="gymForm">

      <!-- Full Name -->

      <div class="form-group">

        <label for="name">Full Name</label>

        <input
          type="text"
          id="name"
          placeholder="Enter your full name"
        >

        <p id="nameMessage" class="message"></p>

      </div>

      <!-- Age -->

      <div class="form-group">

```

```

    <label for="age">Age</label>

    <input
      type="number"
      id="age"
      placeholder="Enter your age"
    >

    <p id="ageMessage" class="message"></p>
  </div>

  <!-- Email -->

  <div class="form-group">

    <label for="email">Email Address</label>

    <input
      type="text"
      id="email"
      placeholder="Enter your email"
    >

    <p id="emailMessage" class="message"></p>
  </div>

  <!-- Phone -->

  <div class="form-group">

    <label for="phone">Phone Number</label>

    <input
      type="text"
      id="phone"
      placeholder="Enter 10-digit phone number"
    >

    <p id="phoneMessage" class="message"></p>
  </div>

  <!-- Gender -->

  <div class="form-group">

    <label>Gender</label>

    <div class="gender-options">

      <label>
        <input
          type="radio"
          name="gender"
          value="Male"
        >
        Male
      </label>

      <label>
        <input
          type="radio"
          name="gender"
          value="Female"
        >
        Female
      </label>

      <label>
        <input
          type="radio"
          name="gender"
          value="Other"
        >

```

```

        Other
    </label>

</div>

<p id="genderMessage" class="message"></p>

</div>

<!-- Membership -->

<div class="form-group">

    <label for="membership">
        Membership Plan
    </label>

    <select id="membership">

        <option value="">
            Select Membership Plan
        </option>

        <option value="Monthly">
            Monthly
        </option>

        <option value="Quarterly">
            Quarterly
        </option>

        <option value="Half-Yearly">
            Half-Yearly
        </option>

        <option value="Yearly">
            Yearly
        </option>

    </select>

    <p id="membershipMessage" class="message"></p>

</div>

<!-- Fitness Goal -->

<div class="form-group">

    <label for="goal">
        Fitness Goal
    </label>

    <select id="goal">

        <option value="">
            Select Fitness Goal
        </option>

        <option value="Weight Loss">
            Weight Loss
        </option>

        <option value="Muscle Gain">
            Muscle Gain
        </option>

        <option value="General Fitness">
            General Fitness
        </option>

        <option value="Strength Training">
            Strength Training
        </option>

    </select>


```

```

        <p id="goalMessage" class="message"></p>

    </div>

    <button type="submit">
        Submit Application
    </button>

</form>

<!-- Dynamic Registration Result -->
<div id="registrationResult">

    <div class="result-header">

        <h2>Registration Successful!</h2>

        <p>Gym Membership Details</p>

    </div>

    <div class="details">

        <div class="detail-row">

            <span class="detail-label">
                Full Name
            </span>

            <span
                id="resultName"
                class="detail-value"
            ></span>

        </div>

        <div class="detail-row">

            <span class="detail-label">
                Age
            </span>

            <span
                id="resultAge"
                class="detail-value"
            ></span>

        </div>

        <div class="detail-row">

            <span class="detail-label">
                Email
            </span>

            <span
                id="resultEmail"
                class="detail-value"
            ></span>

        </div>

        <div class="detail-row">

            <span class="detail-label">
                Phone Number
            </span>

            <span
                id="resultPhone"
                class="detail-value"
            ></span>

        </div>

        <div class="detail-row">

```

```

        <span class="detail-label">
            Gender
        </span>

        <span
            id="resultGender"
            class="detail-value"
        ></span>

    </div>

    <div class="detail-row">

        <span class="detail-label">
            Membership Plan
        </span>

        <span
            id="resultMembership"
            class="detail-value"
        ></span>

    </div>

    <div class="detail-row">

        <span class="detail-label">
            Fitness Goal
        </span>

        <span
            id="resultGoal"
            class="detail-value"
        ></span>

    </div>

    <div class="registered">
        ✓ Member registered successfully!
    </div>

</div>

</div>

<script>

    // =====
    // GET HTML ELEMENTS
    // =====

    var nameInput = document.getElementById("name");
    var ageInput = document.getElementById("age");
    var emailInput = document.getElementById("email");
    var phoneInput = document.getElementById("phone");
    var membershipInput =
        document.getElementById("membership");
    var goalInput =
        document.getElementById("goal");
    var form =
        document.getElementById("gymForm");
    var registrationResult =
        document.getElementById("registrationResult");

    // =====
    // NAME VALIDATION
    // =====

```

```

function validateName() {

    var name = nameInput.value.trim();

    var message =
        document.getElementById("nameMessage");

    var namePattern = /^[A-Za-z ]+$/;

    if (name === "") {

        message.textContent =
            "Name is required.";

        message.className =
            "message error";

        nameInput.classList.add(
            "invalid-input"
        );

        nameInput.classList.remove(
            "valid-input"
        );

        return false;

    }

    else if (!namePattern.test(name)) {

        message.textContent =
            "Name should contain only letters.";

        message.className =
            "message error";

        nameInput.classList.add(
            "invalid-input"
        );

        nameInput.classList.remove(
            "valid-input"
        );

        return false;

    }

    else {

        message.textContent =
            "✓ Name is valid.";

        message.className =
            "message success";

        nameInput.classList.add(
            "valid-input"
        );

        nameInput.classList.remove(
            "invalid-input"
        );

        return true;

    }

}

// =====
// AGE VALIDATION
// =====

function validateAge() {

    var age = ageInput.value;

```

```

var message =
    document.getElementById("ageMessage");

if (age === "") {

    message.textContent =
        "Age is required.";

    message.className =
        "message error";

    ageInput.classList.add(
        "invalid-input"
    );

    ageInput.classList.remove(
        "valid-input"
    );

    return false;

}

else if (age < 16 || age > 60) {

    message.textContent =
        "Age must be between 16 and 60.";

    message.className =
        "message error";

    ageInput.classList.add(
        "invalid-input"
    );

    ageInput.classList.remove(
        "valid-input"
    );

    return false;

}

else {

    message.textContent =
        "✓ Valid age.";

    message.className =
        "message success";

    ageInput.classList.add(
        "valid-input"
    );

    ageInput.classList.remove(
        "invalid-input"
    );

    return true;

}

}

// =====
// EMAIL VALIDATION
// =====

function validateEmail() {

    var email =
        emailInput.value.trim();

    var message =
        document.getElementById("emailMessage");

    var emailPattern =

```

```

    /^[^\s@]+@[^\s@]+\.[^\s@]+$/;

    if (email === "") {

        message.textContent =
            "Email is required.";

        message.className =
            "message error";

        emailInput.classList.add(
            "invalid-input"
        );

        emailInput.classList.remove(
            "valid-input"
        );

        return false;
    }

    else if (!emailPattern.test(email)) {

        message.textContent =
            "Enter a valid email address.";

        message.className =
            "message error";

        emailInput.classList.add(
            "invalid-input"
        );

        emailInput.classList.remove(
            "valid-input"
        );

        return false;
    }

    else {

        message.textContent =
            "✓ Email is valid.";

        message.className =
            "message success";

        emailInput.classList.add(
            "valid-input"
        );

        emailInput.classList.remove(
            "invalid-input"
        );

        return true;
    }
}

// =====
// PHONE VALIDATION
// =====

function validatePhone() {

    var phone =
        phoneInput.value.trim();

    var message =
        document.getElementById("phoneMessage");

    var phonePattern =
        /^[0-9]{10}$/;

```

```

    if (phone === "") {
        message.textContent =
            "Phone number is required.";

        message.className =
            "message error";

        phoneInput.classList.add(
            "invalid-input"
        );

        phoneInput.classList.remove(
            "valid-input"
        );

        return false;
    }

    else if (!phonePattern.test(phone)) {

        message.textContent =
            "Phone number must contain 10 digits.";

        message.className =
            "message error";

        phoneInput.classList.add(
            "invalid-input"
        );

        phoneInput.classList.remove(
            "valid-input"
        );

        return false;
    }

    else {

        message.textContent =
            "✓ Phone number is valid.";

        message.className =
            "message success";

        phoneInput.classList.add(
            "valid-input"
        );

        phoneInput.classList.remove(
            "invalid-input"
        );

        return true;
    }
}

// =====
// GENDER VALIDATION
// =====

function validateGender() {

    var gender =
        document.querySelector(
            'input[name="gender"]:checked'
        );

    var message =
        document.getElementById(
            "genderMessage"
        );

    if (gender == null) {

```

```

        message.textContent =
            "Please select your gender.";

        message.className =
            "message error";

        return false;
    }

    else {

        message.textContent =
            "✓ Gender selected.";

        message.className =
            "message success";

        return true;
    }
}

// =====
// MEMBERSHIP VALIDATION
// =====

function validateMembership() {

    var message =
        document.getElementById(
            "membershipMessage"
        );

    if (membershipInput.value === "") {

        message.textContent =
            "Please select a membership plan.";

        message.className =
            "message error";

        return false;

    }

    else {

        message.textContent =
            "✓ Membership selected.";

        message.className =
            "message success";

        return true;
    }
}

// =====
// FITNESS GOAL VALIDATION
// =====

function validateGoal() {

    var message =
        document.getElementById(
            "goalMessage"
        );

    if (goalInput.value === "") {

        message.textContent =
            "Please select your fitness goal.";

        message.className =
            "message error";
    }
}

```

```

        return false;
    }

    else {

        message.textContent =
            "✓ Fitness goal selected.";

        message.className =
            "message success";

        return true;
    }
}

// =====
// LIVE INPUT CHECKING
// =====

nameInput.addEventListener(
    "input",
    validateName
);

ageInput.addEventListener(
    "input",
    validateAge
);

emailInput.addEventListener(
    "input",
    validateEmail
);

phoneInput.addEventListener(
    "input",
    validatePhone
);

membershipInput.addEventListener(
    "change",
    validateMembership
);

goalInput.addEventListener(
    "change",
    validateGoal
);

// Gender live validation

var genderButtons =
    document.querySelectorAll(
        'input[name="gender"]'
    );

genderButtons.forEach(
    function(gender) {

        gender.addEventListener(
            "change",
            validateGender
        );

    }
);

// =====
// FORM SUBMISSION
// =====

form.addEventListener(
    "submit",
    function(event) {

        event.preventDefault();
    }
);

```

```

// Validate all fields

var validName =
    validateName();

var validAge =
    validateAge();

var validEmail =
    validateEmail();

var validPhone =
    validatePhone();

var validGender =
    validateGender();

var validMembership =
    validateMembership();

var validGoal =
    validateGoal();

// Check whether everything is valid

if (
    validName &&
    validAge &&
    validEmail &&
    validPhone &&
    validGender &&
    validMembership &&
    validGoal
) {

    // =====
    // GET REGISTERED VALUES
    // =====

    var registeredName =
        nameInput.value;

    var registeredAge =
        ageInput.value;

    var registeredEmail =
        emailInput.value;

    var registeredPhone =
        phoneInput.value;

    var registeredGender =
        document.querySelector(
            'input[name="gender"]:checked'
        ).value;

    var registeredMembership =
        membershipInput.value;

    var registeredGoal =
        goalInput.value;

    // =====
    // DOM MANIPULATION
    // DISPLAY REGISTERED INFORMATION
    // =====

    document.getElementById(
        "resultName"
    ).textContent =
        registeredName;

    document.getElementById(
        "resultAge"
    ).textContent =
        registeredAge;

```

```

        document.getElementById(
            "resultEmail"
        ).textContent =
            registeredEmail;

        document.getElementById(
            "resultPhone"
        ).textContent =
            registeredPhone;

        document.getElementById(
            "resultGender"
        ).textContent =
            registeredGender;

        document.getElementById(
            "resultMembership"
        ).textContent =
            registeredMembership;

        document.getElementById(
            "resultGoal"
        ).textContent =
            registeredGoal;

        // Show registration result
        registrationResult.style.display =
            "block";

        // Scroll to result
        registrationResult.scrollIntoView({
            behavior: "smooth"
        });
    }
    else {
        registrationResult.style.display =
            "none";
    }
}
);
</script>
</body>
</html>

```

7. Output of the case study :

Name : Kanak Tembhurne PRN : 24070521290 File path :

[http://127.0.0.1:5500/Practical%2002/Casestudy_08_Kanak Tembhurne 1290.html](http://127.0.0.1:5500/Practical%2002/Casestudy_08_Kanak_Tembhurne_1290.html)

Gym Admission Form
Register today and start your fitness journey

Full Name
Kanak Tembhumne
✓ Name is valid.

Age
20
✓ Valid age.

Email Address
kanaktembhumne@gmail.com
✓ Email is valid.

Phone Number
9175539289
✓ Phone number is valid.

Gender
☐ Male
 ☒ Female
 ☐ Other
 ✓ Gender selected.

Membership Plan
Quarterly
✓ Membership selected.

Fitness Goal
General Fitness
✓ Fitness goal selected.

Submit Application

Registration Successful!
Gym Membership Details

Full Name	Kanak Tembhumne
Age	20
Email	kanaktembhumne@gmail.com
Phone Number	9175539289
Gender	Female
Membership Plan	Quarterly
Fitness Goal	General Fitness

✓ Member registered successfully!

8. Conclusion

Thus, HTML form element access and event-driven validation were successfully demonstrated using JavaScript event listeners (input, blur, change, submit).